

02_REQUIREMENTS.md

TASK 02

REQUIREMENTS ENGINEERING

Status

Mandatory

Owner

Requirements Engineer

Supporting Guides

Project Director

Security Architect

Solution Architect

Principal Engineer

Estimated Time

1 to 2 Days

Dependencies

00_PROJECT_CONTEXT.md

00_ENGINEERING_RULES.md

00_PROJECT_MEMORY.md

01_PROJECT_DIRECTOR.md

02_REQUIREMENTS_ENGINEER.md

01_PROJECT_SETUP.md

Mission

Transform the project vision into a complete Software Requirements Specification (SRS).

Do not write implementation code.

Do not design the database.

Do not design APIs.

Do not design the UI.

Your objective is to define exactly what the system must do before any design or development begins.

Every requirement should be measurable, testable and traceable.

Objectives

Create a complete Software Requirements Specification.

Remove ambiguity.

Identify business rules.

Define security expectations.

Prepare the project for architecture.

Ensure every future engineering task has a reliable foundation.

Required Deliverables

Generate the following documents.

Software Requirements Specification

Stakeholder Analysis

Business Requirements

Functional Requirements

Non Functional Requirements

Security Requirements

Privacy Requirements

Legal Requirements

Business Rules

System Scope

Out of Scope

Assumptions

Constraints

Success Criteria

Acceptance Criteria

Requirements Traceability Matrix

Risk Register

STRIDE Analysis

MoSCoW Prioritisation

Glossary

Revision History

Stakeholder Analysis

Identify every stakeholder.

Administrator

Coach

Parent

Artist

System Administrator

Future Developers

University Assessors

For each stakeholder define

Responsibilities

Goals

Permissions

Pain Points

Security Concerns

Success Metrics

Functional Requirements

Define every system feature.

Authentication

Registration

Login

Logout

Forgot Password

Reset Password

User Management

Role Management

Artist Management

Coach Management

Parent Management

Attendance Management

Performance Management

Injury Management

Medical Records

Emergency Contacts

Activities

Notifications

Reports

Dashboard

Audit Logs

Profile Management

Settings

Search

Filtering

Export

Every requirement should include

Unique Identifier

Description

Business Value

Priority

Dependencies

Acceptance Criteria

Security Notes

Non Functional Requirements

Define measurable requirements.

Security

Performance

Availability

Reliability

Scalability

Maintainability

Accessibility

Usability

Compatibility

Portability

Disaster Recovery

Backup

Logging

Monitoring

Every requirement must include measurable success criteria.

Security Requirements

Define requirements for

Authentication

Authorization

RBAC

Password Policy

Session Management

Audit Logging

Rate Limiting

Validation

Encryption

Secrets Management

Error Handling

OWASP Compliance

NIST Compliance

Every security requirement must include its business justification.

Privacy Requirements

Document

Personal Data

Medical Information

Emergency Contacts

Attendance Records

Performance Records

Child Data Protection

Consent

Retention

Deletion

Access Control

Privacy should influence every design decision.

Legal Requirements

Consider

GDPR

Singapore PDPA

Copyright

Intellectual Property

Privacy

Data Retention

Consent

Record legal obligations clearly.

Business Rules

Document every rule.

Examples

Parents access only their children.

Coaches manage assigned artists.

Attendance cannot be duplicated.

Medical records require authorization.

Deleted users remain in audit history.

Every rule must be enforceable.

User Stories

Create user stories for every feature.

Format

As a

I want

So that

Include

Priority

Acceptance Criteria

Security Considerations

Business Value

Use Cases

Produce complete use cases.

Actor

Preconditions

Main Flow

Alternative Flow

Exception Flow

Postconditions

Security Requirements

Expected Result

Use cases will support later diagrams.

Abuse Cases

Think like an attacker.

Examples

Unauthorized access

Privilege escalation

Session theft

Malicious uploads

Data tampering

Every abuse case must include mitigation.

Misuse Cases

Think about accidental misuse.

Examples

Incorrect attendance

Weak passwords

Duplicate records

Expired sessions

Large uploads

Design requirements should reduce user mistakes.

Personas

Develop realistic personas.

Administrator

Coach

Parent

Adult Artist

Child Artist

Include

Goals

Experience

Frustrations

Technical Skills

Accessibility Needs

User Journey Maps

Map complete workflows.

Registration

Authentication

Attendance

Performance Tracking

Medical Updates

Reports

Logout

Identify

Pain Points

Security Risks

Improvement Opportunities

Risk Register

For every identified risk document

Risk ID

Description

Likelihood

Impact

Priority

Mitigation

Owner

Residual Risk

STRIDE Analysis

Analyse every major feature.

Authentication

Attendance

Medical Records

Reports

Dashboard

Administration

Document

Threat

Likelihood

Impact

Mitigation

Residual Risk

Requirements Traceability Matrix

Every requirement must link to

Business Goal

Feature

Security Requirement

Database

API

UI

Test Case

Formal Method

Report Section

No requirement should exist without traceability.

Prioritisation

Use MoSCoW.

Must Have

Should Have

Could Have

Won't Have

Security requirements always receive Must Have priority.

Deliverables

Software Requirements Specification

Business Requirements Document

Stakeholder Analysis

Functional Requirements Catalogue

Non Functional Requirements Catalogue

Security Requirements Specification

Privacy Requirements
Legal Requirements
Business Rules Catalogue
Use Cases
User Stories
Abuse Cases
Misuse Cases
Personas
Journey Maps
Risk Register
STRIDE Analysis
Requirements Traceability Matrix
Feature Priority Matrix
Glossary

Review Checklist

Verify

Every stakeholder is represented.

Every feature has business value.

Requirements are measurable.

Acceptance criteria exist.

Security requirements are complete.

Privacy requirements are documented.

Legal requirements are addressed.

Business rules are consistent.

No conflicting requirements exist.

Everything is traceable.

Update Project Memory

Mark

Requirements Engineering

Completed

Record

Approved Requirements

Business Rules

Project Scope

Known Risks

Outstanding Decisions

Next Task

System Architecture

Definition of Done

The Software Requirements Specification is complete.

Every requirement is approved.

Security requirements are complete.

Business rules are documented.

Acceptance criteria exist.

Traceability is complete.

Architecture can begin without ambiguity.

No engineering team should need to guess what the system is expected to do.